*Name:*________________________________

ChE 312 Example Exam 1: Closed-book section

30 points, 4 problems, 2 pages. [Our blank page if you need it.]

Write your name on each page, representing your honor commitment.
Be specific about units, watch significant digits, show/explain your calcs. <u>NEVER ERASE</u>!

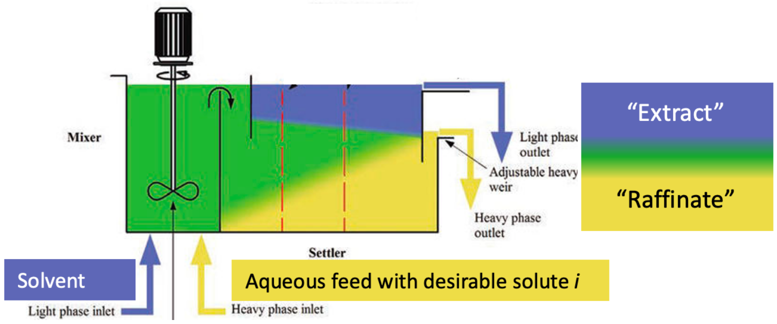
Prob. 1 (10 pts). Define "species volatility" and "relative volatility" for vapor-liquid equilibrium, both in words and with mole-fraction variables. Are these definitions valid for non-ideal gases and liquids?

Prob. 2 (5 pts). To make a distillation column work, what is the purpose of a condenser at the top?

Prob. 3 (5 pts). To make a distillation column work, what is the purpose of a reboiler at the bottom?

Name:______________________________

Prob. 4 (10 pts). A "mixer-settler" for continuous-flow liquid-liquid extraction (see the figure) is just like a flash drum but with two feeds that get mixed together. Two liquids, a lower-density organic solvent ("light phase") and a higher-density aqueous liquid ("heavy phase") are fed and mixed. As the two immiscible phases re-separate, a compound *i* that is soluble in both is transferred to the solvent "extract" phase where it has a higher solubility. The solute-depleted aqueous phase is called the "raffinate." Equilibrium mole fractions for the solute are related by $\gamma_{i,light}x_{i,light} = \gamma_{i,heavy}x_{i,heavy}$.]



Sketch a control volume around the whole mixer-settler; define suitable flow and mole-fraction variables; and write an overall balance and a solute-species balance.

*Name:*________________________________

ChE 312-001 Exam 1, Spring 2023: Open-book section
70 pts, 3 problems (5, 6, 7); 3 pp with graph; extra blank page.

Write your name on each page, representing your honor commitment.
Be specific about units, watch significant digits, show/explain your calcs. <u>NEVER ERASE</u>!

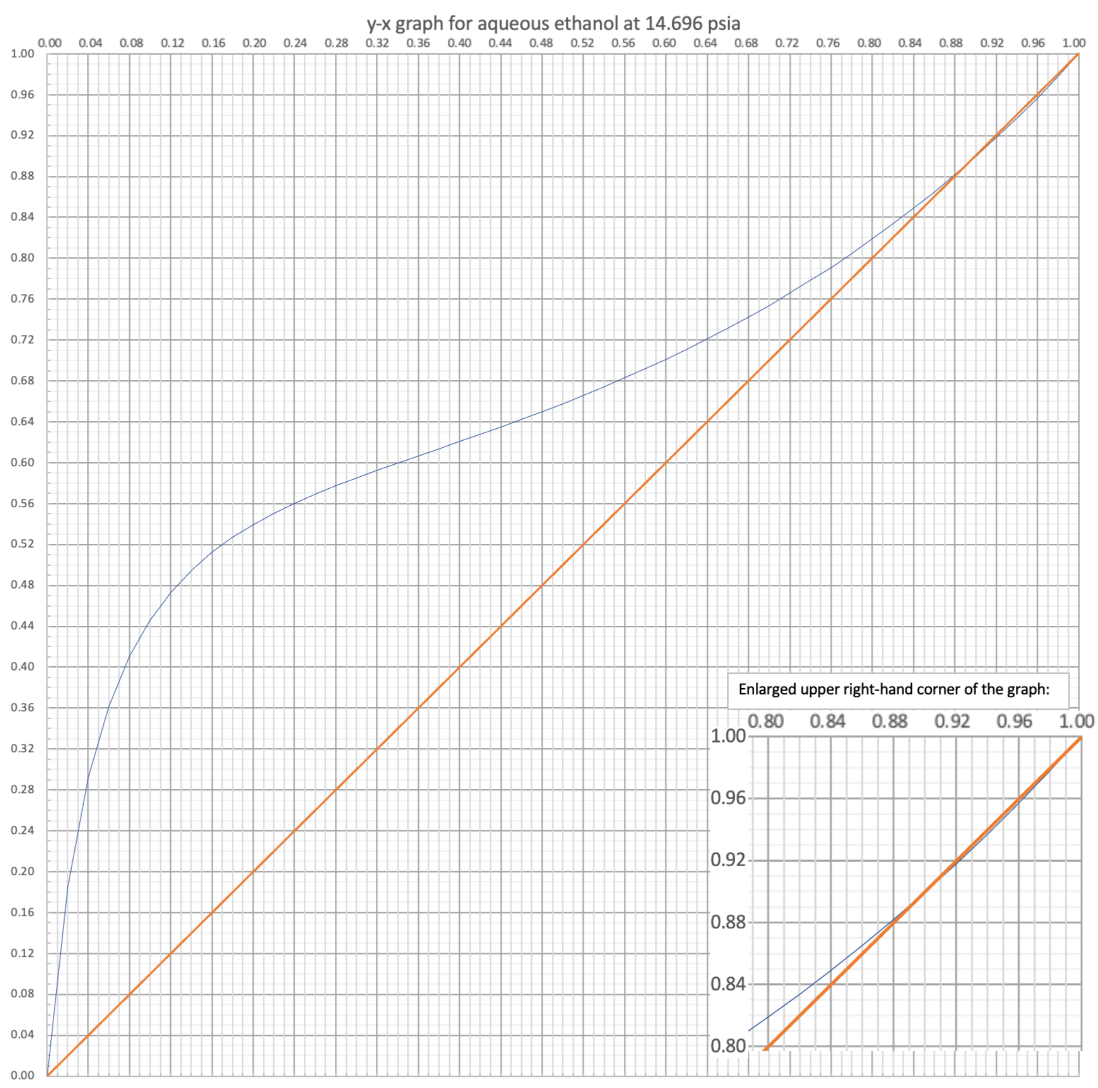
Problem 5 (40 pts).

Purifying bio-based fuels and chemicals is often complicated by their oxygen content. An example is distillation of ethanol-water mixtures, which are nonideal enough to have a minimum-boiling azeotrope at $x_{ethanol}$=0.90. That's okay; we can distill from feeds having $z_{ethanol}$<0.90.

Fermentation ethanol is up to about 12-14 mol% in water. Design a column for 100 mol/h of saturated-liquid feed of $z_{ethanol}$=0.120 to make distillate having $x_{ethanol}$=0.82. There is no specification on the bottoms concentration, but 95% of the feed ethanol should go into the distillate. Equilibrium mole fractions are provided on the attached graph. Use a total condenser and a partial reboiler.

Determine the optimum reflux ratio based on R=1.5 R_{min}; the number of stages; number of trays; and optimal feed stage location. Fill out the table of mole fractions using McCabe-Thiele analysis.

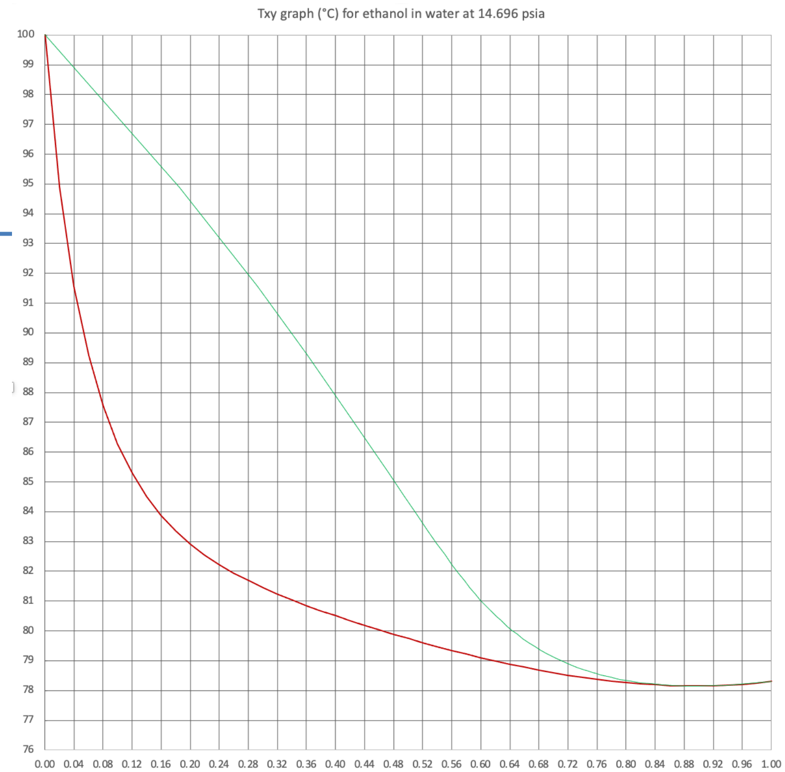
Stage	x	y
1		
2		
3		
4		
5		
6		
7		
8		
9		
10		
11		
12		
13		
14		
15		
16		
17		
18		
19		
20		
21		
22		
23		
24		
25		
26		
27		
28		



*Name:*________________________________

*Name:*__________________________________

Problem 6 (20 pts). Using the Txy equilibrium diagram at right, determine the approximatetemperature at the reboiler, the feed tray, and the top tray. Explain why the temperature changes as it does from the reboiler to the feed tray to the top tray.



Problem 7 (10 pts). The idea of a K_i value in binary or multi-component equilibrium is to compare mole fractions for chemical species i in two equilibrium phases, whether vapor-liquid, liquid-liquid, liquid-solid, vapor-solid, or solid-solid. In equilibrium between an ideal gas mixture and an ideal liquid solution involving multiple compounds, write an equation for K_i for chemical species i in terms of phase mole fractions, pressure, and temperature.

*Name:*__________________________________